ELDER CARE ROBOT PROJECT

PROJECT COST ESTIMATION & BASELINE REPORT

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Timeline: 12-Month Lifecycle

1. Introduction & Core Assumptions

This document aims to provide a transparent overview of the cost structure and financial timeline for the development of the Elder Care Robot and its companion mobile application. The cost estimation process is based on the proposed team structure and technical requirements, governed by the following assumptions:

- The total execution time for the project is strictly estimated at **1 year (12 months)**.
- Standard working hours are set to **160 hours per month** to ensure accurate wage calculations across the lifecycle.
- Hourly rates are defined as 500,000 Tomans for primary full-time employees and 250,000 Tomans for trainees.
- Given the absence of an in-house Hardware Developer, **two specialized contractors at 50% capacity** (matching permanent employee rates) have been added to the team.
- Beyond human resource costs, a dedicated budget of **1.5 Billion Tomans** has been allocated for robotics components, server infrastructure, and marketing campaigns.

2. Cost Estimation Logic (WBS breakdown)

In the cost estimation table, the total budget is structured around a Work Breakdown Structure (WBS) consisting of seven main phases. Team members are allocated in a matrix structure across various activities. Trainees are exclusively present in the first six months, and the marketing team is allocated only in the final three months to align with project financial constraints and requirements.

ID	Task Name (WBS)	Duration	Labor Cost (Tomans)	Materials/Other (Tomans)	Total Cost (Tomans)
0	Project Management	12 months	1,200,000,000	50,000,000	1,250,000,000
1	AI Development	9 months	2,400,000,000	150,000,000	2,550,000,000
2	Hardware Development	9 months	720,000,000	800,000,000	1,520,000,000
3	Software Development	9 months	2,400,000,000	100,000,000	2,500,000,000
4	HW & SW Integration	1.5 months	840,000,000	50,000,000	890,000,000
5	System Testing	1.5 months	840,000,000	50,000,000	890,000,000
6	Pre-Launch Marketing	3 months	480,000,000	300,000,000	780,000,000
-	**Project Total**	**12 months**	**8,880,000,000**	**1,500,000,000**	**10,380,000,000**

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3. Project Cost Baseline

Project cash flow is distributed across a 12-month schedule. The logic governing this distribution is as follows:

- **Months 1 to 6:** Core development phase. This period carries the highest financial burden due to the simultaneous presence of trainees and initial hardware procurement (800 Million Tomans for robotic components).
- **Months 7 to 9:** Trainees complete their tenure, lowering ongoing labor costs. Primary teams focus heavily on finalizing their respective modules.
- **Months 10 to 12:** Technical personnel transition entirely into Integration and System Testing. The marketing team begins their campaign with an escalating budget designed to maximize impact exactly at the product launch.

WBS Component	Mo 1	Mo 2	Mo 3	Mo 4	Mo 5	Mo 6	Mo 7	Mo 8	Mo 9	Mo 10	Mo 11	Mo 12
Project Management	125	125	125	125	125	125	85	85	85	85	80	80
AI Development	305	305	305	305	305	305	240	240	240	0	0	0
Hardware Dev.	280	280	280	280	80	80	80	80	80	0	0	0
Software Dev.	300	290	290	290	290	290	250	250	250	0	0	0
System Integration	0	0	0	0	0	0	0	0	0	610	280	0
Testing Phase	0	0	0	0	0	0	0	0	0	0	280	610
Marketing	0	0	0	0	0	0	0	0	0	210	260	310
Monthly Total	**1,010**	**1,000**	**1,000**	**1,000**	**800**	**800**	**655**	**655**	**655**	**905**	**900**	**1,000**
Cumulative Cost	**1,010**	**2,010**	**3,010**	**4,010**	**4,810**	**5,610**	**6,265**	**6,920**	**7,575**	**8,480**	**9,380**	**10,380**

* All numerical figures in the table above are in **Millions of Tomans**.

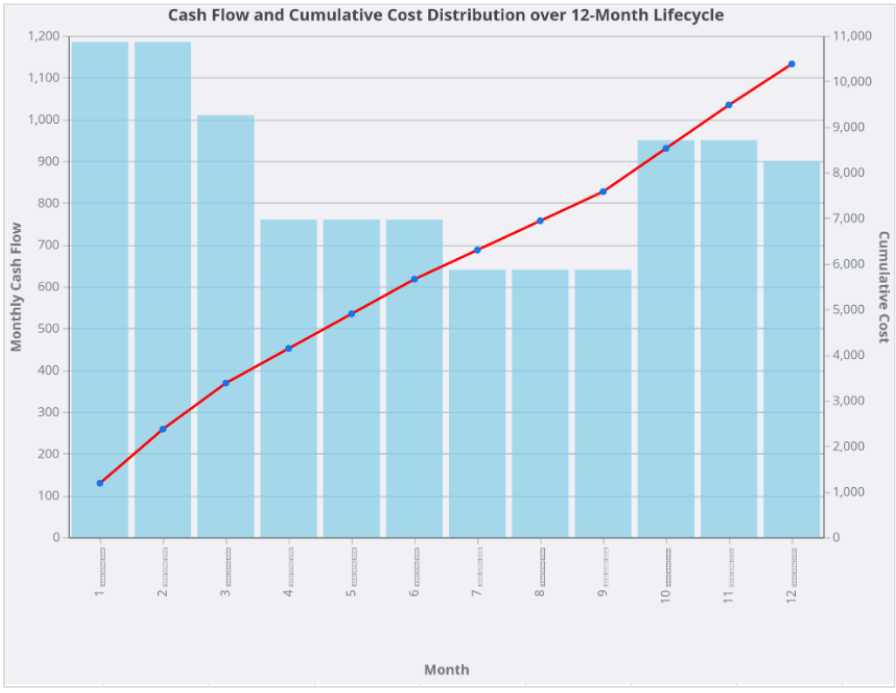


Figure 1: Cash Flow and Cumulative Cost Distribution over the 12-Month Lifecycle

4. Cost Analysis & Economic Justification

Human Resources

Over 85% of the total budget (approx. 8.88 Billion Tomans) is dedicated to recruiting, training, and retaining a specialized technical team. Utilizing trainees in the first 6 months slightly increases immediate costs but creates an internal talent pipeline for future projects.

Hardware Strategy

Addressing the lack of an internal hardware team through partial outsourcing (50% capacity contractors) is a highly efficient mitigation strategy. It prevents the heavy overhead of permanent hires while sufficiently lowering technical delivery risks.

Marketing & Penetration

Concentrating the marketing budget heavily in the final trimester creates a robust demand generation engine (Lead Gen) precisely when the product is ready to ship, thereby drastically accelerating the projected Return on Investment (ROI).

Executive Summary

The total estimated budget required for the successful completion of this 12-month project is **10.38 Billion Tomans**. Considering the immense technical complexity of this product—which demands simultaneous physical manufacturing, AI model training, and cross-platform mobile app development—this estimate is highly reasonable and strongly justified.

Investing in this venture presents exceptional commercialization potential. The "smart elder care" market is currently an untapped niche locally. A product that actively ensures the physical safety of the elderly while providing profound psychological peace of mind for their families possesses extreme intrinsic value. The Cost Baseline reveals that while initial liquidity requirements are high during the hardware procurement phase, the cash burn rate stabilizes significantly in the second half of the project. We recommend that executive management formally approve this budget and authorize the Project Charter to commence operations immediately.

REFERENCES

[1] Project Specification Document (ITPM_Exercise_4.pdf), Faculty of Electrical and Computer Engineering, University of Tehran.

[2] Project Management Institute. *PMBOK® Guide* – 7th Edition (Guidelines for Cost Estimation & Baselines).

[3] Standard IT Service Tariffs & Rate Cards, Iranian ICT Guild Organization.